

Solutions for Homework Assignment #6

Answer to Question 1. We will denote this problem COMPDES, for “compatible design”.

COMPDES is in NP: We prove this using the NTM-based definition of **NP**. A NTM can write the description of a selection of one element from each D_i . This requires $\sum_{i=1}^k \lceil \log_2 |D_i| \rceil$ bits, since we $\log_2 |D_i|$ bits to distinguish the different devices for component i , so it can be done in polynomial time by a NTM. Then the NTM can check that for every one of the $\Theta(k^2)$ pairs of selected devices, it is not in the set X of incompatible pairs. This obviously can be done in polynomial time (deterministically). Therefore there is a polytime NTM that decides COMPDES.

COMPDES is NP-hard: Next we show that COMPDES is **NP**-hard by describing a polytime mapping reduction of CNF-SAT to COMPDES. Given a CNF formula F we will show how to construct an instance $\langle D_1, \dots, D_k, X \rangle$ of COMPDES so that

$$F \text{ is satisfiable if and only if } \langle D_1, \dots, D_k, X \rangle \text{ has a compatible design.} \quad (*)$$

Let $F = C_1 \wedge C_2 \wedge \dots \wedge C_m$ be a CNF formula, where $C_i = \ell_i^1 \vee \ell_i^2 \vee \dots \vee \ell_i^{t_i}$ and each ℓ_i^j is literal. We say that two literals ℓ and ℓ' are **opposite** if one is x and the other is $\bar{x}$ for the same variable x . We can assume, without loss of generality, that the same variable does not appear in the same clause: If there are multiple occurrences of a variable in a clause and all are positive or all are negative, we can delete all occurrences of the variable but one: the new formula is equivalent to the original. If there are both positive and negative occurrences of a variable in a clause, we can delete the entire clause: the new formula is satisfiable if and only if the original one is.

For each $i \in [1..m]$ we let $D_i = \{\ell_i^1, \ell_i^2, \dots, \ell_i^{t_i}\}$ (i.e., the set of “devices” for component i is the literals of the i -th clause). We also let $X = \{\{\ell_i^j, \ell_{i'}^{j'}\} : i \neq i' \text{ and } \ell_i^j \text{ and } \ell_{i'}^{j'} \text{ are opposite literals}\}$. The instance of COMPDES to which we map F is $\langle D_1, D_2, \dots, D_m, X \rangle$.

The size of this is polynomial in the size of F , so it can be constructed in polynomial time. It remains to show (*).

[ONLY IF] Suppose F is satisfiable, and let τ be a truth assignment that satisfies it. Thus, each clause C_i has some literal $\ell_i^{j_i}$ such that $\tau(\ell_i^{j_i}) = 1$; if there are multiple literals that satisfy C_i just pick one of them arbitrarily as $\ell_i^{j_i}$. We claim that $(\ell_1^{j_1}, \ell_2^{j_2}, \dots, \ell_m^{j_m})$ form a compatible design for $\langle D_1, D_2, \dots, D_m, X \rangle$. This is because no two of them are in X : If they were, by definition of X , they would be opposite literals, and therefore τ could not satisfy both of them.

[IF] Suppose $\langle D_1, D_2, \dots, D_m, X \rangle$ has a compatible design $(\ell_1^{j_1}, \ell_2^{j_2}, \dots, \ell_m^{j_m})$. For any variable x of F , define

$$\tau(x) = \begin{cases} 1, & \text{if for some } i \in [1..m], \ell_i^{j_i} = x \\ 0, & \text{if for some } i \in [1..m], \ell_i^{j_i} = \bar{x} \\ 0 \text{ or } 1, \text{ arbitrarily,} & \text{otherwise (i.e., none of the literals in the compatible design are } x \text{ or } \bar{x}) \end{cases}$$

This is well-defined because opposite literals (a) do not appear in the same clause (by assumption), and (b) if they appear in different clauses the pair is in X and therefore the two opposite literals do not appear in the compatible design. By definition, τ satisfies one literal in each clause, and so F is satisfiable.

Answer to Question 2.

FRAMETILING is in NP: We prove this using the certificate-verifier characterization of **NP**. Recall hint (1) regarding how to specify the position of a rectangle on the Cartesian plane. Suppose the frame with

dimensions $A \times B$ is positioned so that its lower-left corner is at the origin $(0, 0)$ and its upper-right corner is at (A, B) . Let (x_{iL}, y_{iL}) and (x_{iR}, y_{iR}) be the positions of the lower-left and upper-right corner of tile i , respectively.

The certificate for a yes-instance of FRAME TILING is a specification of the positions of the given tiles, i.e., a sequence of $2n$ pairs of natural numbers: (x_{iL}, y_{iL}) and (x_{iR}, y_{iR}) , for $i \in [1..n]$. This is a sequence of length $2n$, each element of which requires ℓ bits, where ℓ is the number of bits required for the longest dimension of any tile. So the size of the certificate is a polynomial in the size of the input $\langle F, T \rangle$.

To verify that this is a legitimate certificate for the input, i.e., that the tiles positioned according to the certificate cover precisely the frame (with no spillover, gaps, or overlaps), the verifier checks that:

- (a) every tile's position matches its dimensions placed horizontally or vertically: either (i) $x_{iR} - x_{iL} = a_i$ and $y_{iR} - y_{iL} = b_i$, or (ii) $x_{iR} - x_{iL} = b_i$ and $y_{iR} - y_{iL} = a_i$;
- (b) all tile positions are within the frame (no spillover): $0 \leq x_{iL} < x_{iR} \leq A$ and $0 \leq y_{iL} \leq y_{iR} \leq B$ for each $i \in [1..n]$;
- (c) the area of the frame is equal to the sum of the areas of the tiles (no gaps): $A \cdot B = \sum_{i=1}^n a_i \cdot b_i$; and
- (d) there is no overlap between any two tiles $i, j \in [1..n]$, $i \neq j$: either (i) $x_{iR} \leq x_{jL}$ or $x_{jR} \leq x_{iL}$ (one of i and j is to the left of the other), or (ii) $y_{iL} \leq y_{jR}$ or $y_{jL} \leq y_{iR}$ (one of i and j is above the other)

Each of (a), (b), and (c) can be time in $O(n)$ time, and (d) can be done in $O(n^2)$ time on a random access computer under the unit cost criterion, so the work of the verifier can be done by a deterministic TM in polynomial time.

FRAME TILING is NP-hard: We prove this by demonstrating a polynomial-time mapping reduction of PARTITION to FRAME TILING.

Let $\langle A \rangle$ be an instance of PARTITION, where A is a sequence of positive integers. Given $\langle A \rangle$ we show how to construct (in time polynomial in its size) an instance $\langle F, T \rangle$ of FRAME TILING such that

$$\begin{aligned} & A \text{ can be partitioned into two subsequences each adding up to the same sum} \\ \Leftrightarrow & \text{ Frame } F \text{ can be covered precisely by the tiles in } T \end{aligned} \tag{1}$$

Given the sequence of positive integers $A = a_1, \dots, a_n$ we define the frame F and the sequence of tiles T as follows: $T = (1, 3a_1), (1, 3a_2), \dots, (1, 3a_n)$ — i.e., tile i is 1 unit tall and $3a_i$ units wide; and $F = (2, (3 \cdot \sum_{i \in [1..n]} a_i)/2)$ — i.e., the frame is 2 units tall, and half the sum of the widths of the tiles wide. The intuition behind this construction is that, if A is partitionable into two equal halves, the tiles that correspond to the numbers that form one side of the partition will fill one row of the frame, and the remaining tiles will fill the second row. The proof of (1) given below explains the reason for tripling the elements of A to obtain the width of the tiles.

It is obvious that the size of $\langle F, T \rangle$ is polynomial in the size of $\langle A \rangle$, and that it can be constructed from $\langle A \rangle$ in polynomial time. It remains to prove (1).

[ONLY IF:] Suppose A is partitionable into two equal halves, and let B and C be two subsequences into which A can be partitioned so that the elements of each add up to the same number; that is, $\sum_{b \in B} b = \sum_{c \in C} c = (\sum_{a \in A} a)/2$. Then we get a tiling of the frame by putting the tiles that correspond to the numbers in B in one row of the frame and the tiles that correspond to the numbers in C in the other row of the frame.

[IF:] Suppose a tiling of the frame exists. Since the height of the frame is 2 and the minimum width of a tile is 3, every tile must be positioned horizontally (long side parallel to the x -axis). Since the frame has width $(3 \cdot \sum_{i \in [1..n]} a_i)/2$, the sum of the widths of the tiles in each of the two rows is exactly that. Therefore the sum of the numbers in A that correspond to the tiles in each row is $(\sum_{i \in [1..n]} a_i)/2$, which means that A is partitionable into two equal halves.